IPCON® PET CF

Carbon Fiber Reinforced Polyethylene terephthalate Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.05 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	320 m ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.26 g/cm³
Shrinkage Rate*	Specially-defined	0.17% - 0.46%
Surface Hardness	Shore hardness	86D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	0.38%
Melt Index	270 °C, 2.16 kg	17.8 g/10 min
Melting Temperature	DSC, 10 °C/min	258 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	Not annealed: 142 °C Annealed: 158 °C*
Heat Deflection Temperature	ISO 75, 0.45 MPa	Not annealed: 138 °C Annealed: 155 °C*
Heat Deflection Temperature	ISO 75, 1.80 MPa	Not annealed: 76 °C Annealed: 87 °C*

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	60 MPa
Tensile Strength Z	ISO 527, GB/T 1040	32 MPa
Young's Modulus XY	ISO 527, GB/T 1040	4587 MPa
Young's Modulus Z	ISO 527, GB/T 1040	2832 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	12.4%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	13.1%
Flexural Strength XY	ISO 178, GB/T 9341	134 MPa
Flexural Strength Z	ISO 178, GB/T 9341	68 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	5162 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	3145 MPa

Impact Strength XY	ISO 179, GB/T 1043	22.5 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	8.4 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 290 °C, bed temperature: 90 °C, printing speed: XY—100 mm/s, Z—98 mm/s) and IPCON PET CF (Black) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The XY specimens' annealing parameter is: 80 °C，4 h → 90 °C，8 h. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. High strength, high stiffness, higher toughness and layer adhesion strength, load-bearing and compression resistance; low warping, creep resistance, higher dimensional accuracy; water and oil resistance, upgraded weather resistance, long-term use in indoor water environments and outdoors; easy to print, compatible with AMS and CFS, and supports high speed (maximum speed of more than 300 mm/s).
2. Functional prototypes, tooling fixtures, power tool parts, electronic and electrical casings, high-pressure water gun casings, water pipe connectors, tool box handles, etc.

• Recommended Printing Settings

Drying Settings Before Printing	80 - 110 °C, 6 - 10 h (Blast drying oven)
Printing Temperature and Humidity	≤ 45 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PET
Compatible Printer Type	Enclosed-frame (Recommended), open-frame,
AMS or CFS	Compatible
Compatible Nozzle Material	Hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.4 mm / 0.6 mm (Recommended) / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick
Nozzle Temperature	265 - 295 °C
Bed Temperature	70 - 90 °C
Fan Speed	10% - 40%
Max Print Speed	300 mm/s (P1S, 290 °C)

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